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REVIEW

Changes When the Brain Is Stimulated by Current

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Transcranial electric current stimulations, including transcranial direct, alternating, and random noise stimulation, have been employed to change neuronal membrane potential and modulate firing rates, or facilitate/inhibit ongoing rhythms in the cortex. We review recent findings in the literature of transcranial electric current stimulations with the specific aim of demonstrating how each of these techniques changes the state of the brain, and specifically suggesting that the multiscale entropies and the long-distance synchronies of distributed neural activities in the brain may be modulated by these transcranial electric current stimulations. Moreover, we discuss the possibility of using different combination of electrical current type and electrode montage to further boost the performance of cognitive functions.

KEYWORDS: tDCS, tACS, tRNS, Multiscale Entropy, Complexity, Plasticity, Brain Stimulation.

CONTENTS

In the past decade, many key mechanisms of cognitive neural processes have been revealed by two non-invasive techniques of brain stimulation: transcranial magnetic stimulation (TMS), which activates axons with short magnetic pulses to generate new action potentials; and transcranial electric current stimulations, including transcranial direct (tDCS), alternating (tACS), and random noise stimulation (tRNS), which changes neuronal membrane potential and modulate firing rates (tDCS and tRNS), or facilitate/inhibit ongoing rhythms in the cortex (tACS). This review focuses on tDCS, tACS, and tRNS.

The current study reviews recent findings of transcranial electric current stimulations with the specific aim of demonstrating how these techniques change the state of the brain. To this end, when applying brain stimulation, we suggest that two questions should be kept in mind. Firstly, *what is the key facet of the brain "state" that dominates the cognitive function we are exploring*? Although the states of single neuron have been investigated by prior research, the definition of the state of the whole brain remains unclear. Some certain cognitive processes have been identified to be related to local or long-range coupling of inhibition interneurons (e.g., Ref. [1]). Obviously, the firing rate, oscillatory phase, interneuron coupling, and neurotransmitter etc., each is one facet of the state of the brain (see also Ref. [2]).

Till now, the best we can do is to identify the essential key facet of the brain state corresponding to a certain cognitive function. Secondly, *how do we change the state of the brain*? Transcranial magnetic or electric methods are important because each of them can noninvasively change the performance of the brain in certain cognitive functions. But what is the actual change that is introduced to the brain by these stimulations? In addition, if we have known the key mechanism or key modulations (e.g., phase synchronization, complexity) in a certain cognitive function, can we "inject" these factors into the brain using an appropriate electromagnetic stimulation to change the state of the brain?

CHANGES IN PLASTICITY USING tDCS OR tRNS

Neuronal activity is characterized by the frequency of spike firing, and is controlled by the neuronal membrane potential. While more positive potentials give rise to increased discharge rates, more negative potentials result in reduced firing rates [3]. tDCS is a neurotechnology that injects small direct currents to the surface of the scalp to elicit modulation of neural activities, and has been demonstrated as an efficient tool to selectively modulate these neuronal activities. Depending on the current polarity, duration, and strength, it can induce after-effect excitability changes in the human cortex [4–6]. In addition to the discharge rates, prior research has indicated that tDCS can lead to a change in *N*-methyl-*D*-aspartate (NMDA)-receptor activation. Based on these physiological

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effects, tDCS has the ability to positively affect the short-term cognitive performance (e.g., in implicit motor learning or probabilistic learning [7, 8]; in language learning [9]; in inhibitory control [10, 11]; in memory of digits and words [12–15]; in visual working memory [16]) and improve the rehabilitation potential of neurologic patients (e.g., in chronic aphasia [17]; in AD [18]; for a review, see Ref. [19]).

In contrast to tDCS, tRNS [20–22] is not sensitive to the current polarity and includes a normally distributed random level of current in the KHz range without overall DC offset. The tRNS current, in the frequency spectrum, has a similar pattern with a "white noise" characteristic. tRNS can induce an increase of excitability through both physiological measures and behavioral tasks [23]. Prior research proposed that this facilitatory effect may be due to the repeated opening of sodium channels or to a higher sensitivity of neuronal networks to field modulation than the average single neuron threshold [24]. A recent research [25] indicated that tRNS facilitated task performance only when it was applied during task execution (online), whereas anodal tDCS induced a larger facilitation if it was applied before task execution (offline).

PERSPECTIVE OF MULTISCALE ENTROPY

Instead of using neurophysiological indexes to explain the effect caused by tDCS/tRNS, here we introduce a "*physical*" index—multiscale entropy (MSE) [26] to interpret the physiological effects of tDCS/tRNS. MSE, in short, is an extension of Shannon's entropy [27] and Pincus' approximation entropy [28]. The MSE calculates sample entropies for electrophysiological signals by grouping data points with different timescales (coarse-grained time series). Sample entropy of each coarse-grained time series is a measure to quantify signal complexity by evaluating the occurrence of repetitive patterns. Therefore, low MSE signifies that the time series is more deterministic or regular, whereas high MSE indicates that the signal is more complex and information rich. Prior research has indicated that physical and mental illness are often related to decreased MSE in physiological signals (e.g., Refs. [26, 29, 30]). In addition, since maturation appears to lead to a brain with greater functional variability, which is indicative of enhanced neural complexity, increase of MSE in brain signals has also been demonstrated as a critical index in development [31]. For clinical application, normal subjects and psychiatric patients have been shown to possess electrophysiological signals that differ in degrees of MSE complexity [32]. There is also a well-articulated theoretical framework that emphasizes the space-time structure revealed by MSE as vital to understanding brain mechanisms of cognition and behavior [33, 34]. Therefore, the current employment of MSE to interpret the tDCS/tRNS effect is motivated by three basic hypotheses:

(1) the complexity of a biological system reflects its ability to adapt and function in an ever-changing environment;
(2) biological systems need to operate across multiple spatial and temporal scales, and hence their complexity is also multi-scaled;
(3) the "ability to adapt" in the brain for a cognitive function is associated with the neuroplasticity of this function.

Thus, transcranial electric current stimulations that can efficiently change the plastic excitability in cognitive

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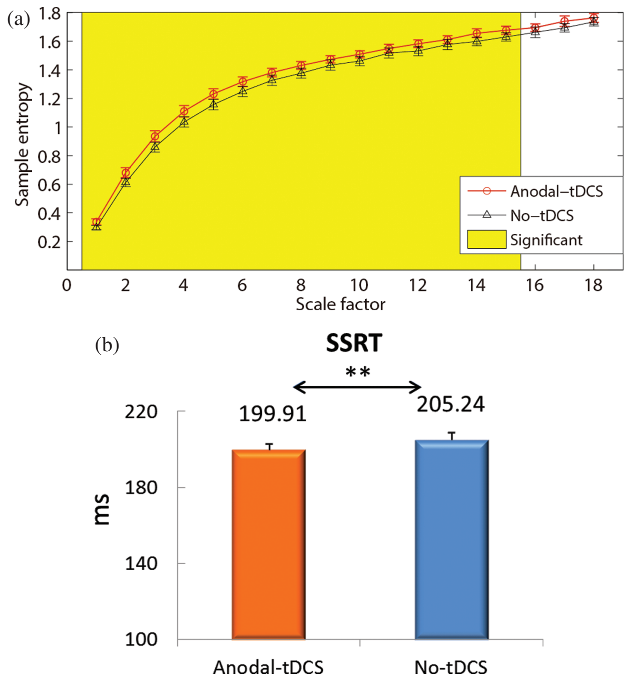

Fig. 1. (A) Hypothetical graph of the complexity elevation by anodal tDCS. MSE of EEG signals over the preSMA brain area is higher in the Anodal-tDCS condition (red line) than the No-tDCS condition (black line). The yellow region indicates that in these scales the sample entropies are speculated to be significantly higher after anodal tDCS. (B) Performance facilitation by anodal tDCS. The stop-signal reaction time (SSRT) of stop trials were significantly reduced by anodal tDCS (paired t-test, $t(15) = 3.069$, $P = .008$); Adapted with permission from [10], T. Y. Hsu, et al., Modulating inhibitory control with direct current stimulation of the superior medial frontal cortex. *Neuroimage* 56, 2249 **(2011)**. © 2011.

functions may also show an increase/decrease of complexity in their brain signals. In a tDCS (over preSMA) study of inhibitory control with a stop-signal task, we have confirmed the facilitatory and inhibitory effect on behavioral performance (inhibition function) from anodal tDCS and cathodal tDCS, respectively [10]. This serves as a suitable platform to test whether MSE would be modulated by tDCS in a way that corresponds to the behavioral findings. According to previous MSE findings in other domains, we speculate that tDCS would significantly affect the MSE of executive function (see Fig. 1(a) for a hypothetical graph).

MANIPULATE SYNCHRONIZATION USING tACS

tACS is a protocol of current stimulation that allows manipulation of intrinsic cortical oscillations with externally applied electrical oscillatory stimulations. tACS is capable of entraining brain oscillations and modulating brain activity in a frequency- and topographic-specific manner (e.g., Refs. [35–39]). It has been proposed that our brain uses long-distance synchronies of distributed neural activities to perform neural communications and to support plasticity (e.g., Refs. [40–43]). The idea of coherence or neural synchronization over a large distance can be thought via the analogy of people trying to overcome jet-lag by adjusting their circadian rhythm according to the new time zone [44]. Similar processes of rhythmic adjustments also take place within a network of different brain areas in order to achieve event-related phase synchronization. In the living brain, the periodic patterning of these responses comes from oscillations generated within the various pools of inhibitory interneurons (e.g., Ref. [45]). These interneurons are coupled through chemical and electrical synapses and are able to sustain local (~ 1 cm through monosynaptic connections with conduction delays of 4–6 ms) and long-range (> 1 cm over polysynaptic pathways with transmission delays longer than 8–10 ms) oscillatory activity patterns [46]. A 2012 study [47] has demonstrated that tACS applied synchronously (0°) over left prefrontal and parietal cortices at 6 Hz significantly improved reaction times in a visual memory-matching task, whereas this tACS applied desynchronously (180°) impaired performance in this task. Therefore, if a procedure of long-distance synchronization amongst brain areas is essential for a certain cognitive process, tACS with an adequate frequency may be employed to facilitate or impair the performance of this cognitive process.

DELIVER FOCAL CURRENT STIMULATION USING MULTIPLE ELECTRODES

The present paradigm of tDCS/tACS/tRNS uses two relatively large electrodes to inject current through the head resulting in electric fields that are broadly distributed over large regions of the brain. A 2011 study [48] proposed a method that used multiple small electrodes (i.e., 1.2 cm diameter) and systematically optimized the applied currents to achieve effective and focal stimulation while ensuring safety of stimulation. This method employed precise forward models of current flow that are similar to that used to solve electroencephalography (EEG) inverse problems. This results in a linear system relating the distribution of the scalp currents to the electrical field, and allows for achieving desired field strength at the target by superposing the field produced by each electrode pair linearly. Therefore, if a focal brain region has been identified (e.g., by fMRI analysis) to be essential for a certain cognitive function, this method may be employed to stimulate the small brain region while reducing the influence on nearby areas. For example, we may apply this multi-electrode current stimulation to a focal essential brain region identified by functional magnetic resonance imaging (fMRI) to boost or impair executive function more efficiently.

CONCLUSION

In the perspective of this review is that we should know the key modulations in the neural mechanism of a certain cognitive function before stimulating our brain by current. A suggested practice is performing time-frequency analysis (e.g., power spectra, coherence, phase-locking value), MSE analysis, fMRI analysis, or dynamical causal modeling (DCM) to better understand the key modulations before the brain stimulation. Here we list the MSE, long-range synchronization, and the functional related brain region focality as indexes to determine which transcranial current stimulation protocol may be efficient for a certain cognitive process. Moreover, since these protocols are not mutually independent and are formed by their specific combination of electrical current type and electrode montage, the challenge in the future work may be how to employ various combinations of current type and electrode montage to further boost the performance of cognitive functions. For example, we may apply tRNS in conjunction with the multi-electrode montage to focally deliver random noise current to a small brain region to test whether this region is really critical for a certain cognitive function. These new protocols together with analyzing the brain signal with informative measures, like MSE, may shed more light on essential neural mechanisms of cognitive functions, and offer the theoretical base of clinical intervention by transcranial current stimulation.

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